

# Vector Calculus

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## Introduction

These notes are composed from textbooks **Erwin Kreyszig, Advanced Engineering Mathematics**.

## 1 Vectors in 2-space and 3-space

*3-space is three dimensional space and 2-space is the plane(special case of 3-space)*

**scalars** are numbers with magnitude only and **vectors** are arrows having specified length(magnitude) pointing in certain direction.

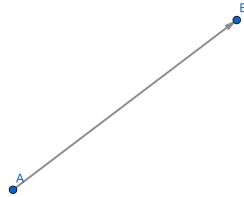


Figure 1: Vector

here, **A** is called the initial point(tail) and **B** is called the terminal point(tip).

Vectors are used in translation(displacement without rotation).  
Two vectors  $\vec{a}$  and  $\vec{b}$  are said to be **equal**, if they have same length and direction.

**Magnitude** of  $\vec{a}$  is written as  $|\vec{a}|$  (**Euclidean Norm**). Unit vector has norm of 1.

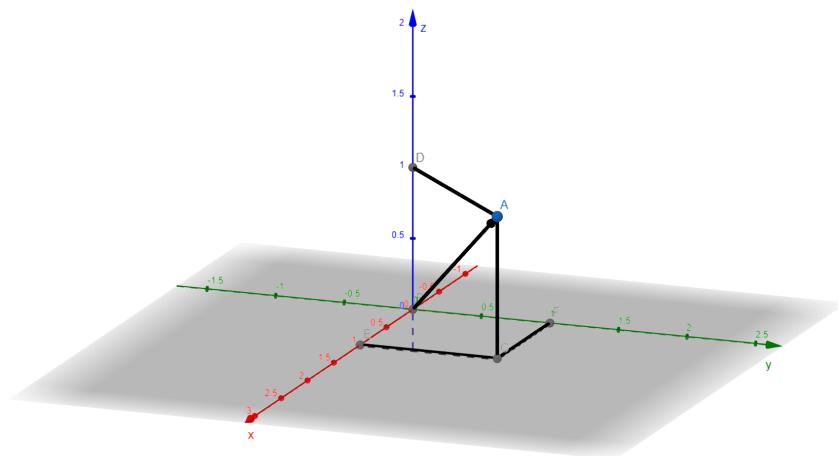


Figure 2: 3D vector

This figure shows the **position vector**  $\vec{r} = (1, 1, 1)$ . Position vector means vector with the tail as origin and tip as some point in  $\mathbb{R}^3$ .

Imagine two points in 3-space,  $P(x_1, y_1, z_1)$  and  $Q(x_2, y_2, z_2)$  then components of  $\vec{PQ}$  are  $a_1 = x_2 - x_1, a_2 = y_2 - y_1, a_3 = z_2 - z_1$ .  $\vec{PQ} = \vec{a} = [a_1, a_2, a_3]$ .  $|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$  (**3D Pythagorus Theorem**)

## 1.1 Vector Addition(Resultant Vector)

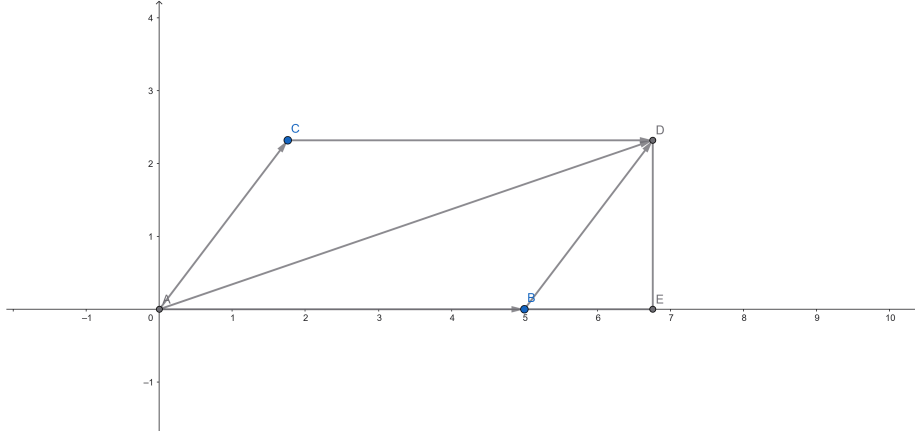


Figure 3: Resultant Vector

In this figure, we consider two vectors  $\vec{AB}$  and  $\vec{AC}$  with the angle between them being  $\angle CAB = \theta$ . Now, to find the **resultant** of these two vectors, we use **parallelogram law of vector addition**. This law basically says, if two vectors are thought of two adjacent sides of parallelogram, it's diagonal is the resultant. In this case diagonal ( $\vec{AD}$ ) is resultant of  $\vec{AB} + \vec{AC}$ .

Now, let  $\alpha$  be the angle made by resultant with the positive x-axis(horizontal in anticlockwise direction).

Drop  $DE \perp$  x-axis,  $\angle DBE = \angle CAB = \theta$  and  $BD = AC$ . Thus,  $BE = BD \cos \theta$  and  $DE = BD \sin \theta$ .

$$\begin{aligned} AD^2 &= (AB + BE)^2 + DE^2 \\ &= (AB + BD \cos \theta)^2 + (BD \sin \theta)^2 \\ &= AB^2 + BD^2 \cos^2 \theta + BD^2 \sin^2 \theta + 2 \cdot AB \cdot BD \cdot \cos \theta \\ &= AB^2 + AC^2 + 2 \cdot AB \cdot AC \cdot \cos \theta \end{aligned}$$

$$\tan \alpha = \frac{b \sin \theta}{a + b \cos \theta}$$

### Some Basic Properties of Vector Addition:

$$\begin{aligned} \vec{a} + \vec{b} &= \vec{b} + \vec{a} \\ (\vec{u} + \vec{v}) + \vec{w} &= \vec{u} + (\vec{v} + \vec{w}) \\ \vec{a} + \vec{0} &= \vec{0} + \vec{a} = \vec{a} \\ \vec{a} + (-\vec{a}) &= \vec{0} \end{aligned}$$

## 1.2 Scalar Multiplication (Multiplication by numbers)

$c\vec{a} = \{ca_1, ca_2, ca_3\}$ , where  $\vec{a} = \{a_1, a_2, a_3\}$

If  $\vec{a} \neq \vec{0}$ , then  $c\vec{a}$  with  $c > 0$  has direction of  $\vec{a}$ , and for  $c < 0$ ,  $c\vec{a}$  has opposite direction.

In all cases,  $|c\vec{a}| = |c||\vec{a}|$  and  $c\vec{a} = \vec{0}$  if  $\vec{a} = \vec{0}$  or  $c = 0$  (or both).

### Some Basic Properties of Scalar Multiplication:

$$c(\vec{a} + \vec{b}) = c\vec{a} + c\vec{b}$$

$$(c + k)\vec{a} = c\vec{a} + k\vec{a}$$

$$c(k\vec{a}) = (ck)\vec{a}$$

$$1\vec{a} = \vec{a}$$

$$0\vec{a} = \vec{0}$$

$$(-1)\vec{a} = -\vec{a}$$

For Unit vectors  $\vec{i}, \vec{j}$  and  $\vec{k}, \vec{a} = \{a_1, a_2, a_3\} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}, \vec{i} = [1, 0, 0], \vec{j} = [0, 1, 0]$  and  $\vec{k} = [0, 0, 1]. \mathbb{R}^3$  has dimension 3.  $\vec{i}, \vec{j}, \vec{k}$  is called a **standard basis** of  $\mathbb{R}^3$ .